Denton, TX, USA
+1 (618) 434 7192
turjakundu@my.unt.edu
turjakundu.github.io

Turja Kundu

Research Focus

High-performance computing; Scalable graph learning; Temporal/dynamic graph learning.

Reference — Dr. Sanjukta Bhowmick (Advisor) `Sanjukta.Bhowmick@unt.edu`; Dr. Yuede Ji `yuede.ji@uta.edu`

Education

2021–Present — **Ph.D. Candidate, Computer Science and Engineering**, *University of North Texas*, Initiated at SIUC

2007–2012 — **B.Sc. in Computer Science and Engineering**, *Bangladesh Univ. of Engineering and Technology (BUET)*

Internship

Summer 2026 — **Computational Science Intern**, *Fermi National Accelerator Laboratory (Fermilab)*, Batavia, IL
Integrated the DUNE CVN classifier with a remote NVIDIA Triton server inside LArSoft/dunereco, and benchmarked CVN and DNN ROI inference as a service.

Research Experience and Publications

ATLAS (Under Review) — **ATLAS: Adaptive Topology-based Learning at Scale for Homophilic and Heterophilic Graphs. T. Kundu**, S. Bhowmick. Under review, KDD. `https://arxiv.org/abs/2512.14908`
Built a propagation-free, topology-augmented MLP: multi-resolution community structure, one-hop neighbor features, and multi-hop label priors are precomputed once and concatenated to node attributes, giving i.i.d. minibatch training and adjacency-free inference. Derived an information-versus-estimation-cost theory predicting which channel and resolution help on a given graph. *Results (19 benchmarks):* best average rank (5.44) among 17 methods on eight medium-scale graphs, tied-best rank (4.40) among scalable baselines on five large-scale graphs, and best rank (1.33 vs. 1.67 for LINKX) on the six million-node LINKX benchmarks; inference on ogbn-products is over 150× faster than ClusterGCN and GraphSAINT.

HIOS (IEEE Cluster 2023) — **HIOS: Hierarchical Inter-Operator Scheduler for Real-Time Inference of DAG-Structured Deep Learning Models on Multiple GPUs. T. Kundu**, T. Shu. IEEE Cluster 2023.
Designed HIOS, a polynomial-time scheduler combining inter-GPU operator mapping by iterative longest-path selection with sliding-window intra-GPU parallelization, implemented on cuDNN and CUDA-aware MPI. *Results:* Improve inference latency by up to 16.5% over the state-of-the-art IOS scheduler for Inception-V3 and NASNet models.

Centrality Resilience (ICCS 2025) — **Centrality Resilience in Complex Networks.** F. A. Irany, **T. Kundu**, S. Sarakar, A. Mukherjee, S. Bhowmick. ICCS 2025, Short Paper.
Studied how well closeness and betweenness rankings are preserved under attack, designing sampling-based edge-attack models that disrupt the rank of the top-k (k=20) high-centrality vertices across ten real-world networks.

Workshop — Y. Zhang, D. Pandey, D. Wu, **T. Kundu**, R. Li, T. Shu. *Accuracy-Constrained Efficiency Optimization and GPU Profiling of CNN Inference.* SC Workshops 2023.

Poster — **T. Kundu**, S. Bhowmick. *Effect of Graph Structure vs. Feature Quality for GCN.* SIAM 2025.

Teaching and Mentoring Experience

TA — Southern Illinois University Carbondale; University of North Texas, 2021–2024. Courses: Linux Programming; Computer Organization/Architecture; Distributed Programming; Advanced OOP; Operating Systems; Graph Theory; Algorithms; Discovering Computer Science.

Coursework and Technical Skills

Coursework — Deep Learning; Distributed Systems; Numerical Computation; Graph Theory; Software–Hardware Codesign for DNN; Computer Architecture; Computational Statistics.

Skills — C, C++, CUDA, MPI, OpenMP, Python, Java, Fortran, R, Bash; PyTorch, cuDNN, NCCL, CUPTI, ROS, NetworkX, scikit-learn, NumPy, Pandas; Nsight Systems/Compute, `perf`, Wireshark.

Awards

NSF Travel Grant, IEEE Cluster 2023; Dean's List Award, BUET, 2010.

Industry Experience

2017–2021 — **Assistant Director (IT)**, ASA International Management Services Ltd, Dhaka – Developed web-based microfinance banking systems and led large-scale data migration.

2015–2017 — **Software Engineer**, Relisource Technology Ltd, Dhaka – Built distributed service-oriented systems using Windows Services.

2012–2015 — **Software Engineer**, Mir Technology Ltd, Dhaka – Developed SIP relay/proxy servers using pthreads/BSD sockets and Android SIP clients.